Taylor Polynomial

nth degree Taylor polynomial of $f(x)$ is $P_n(x) = f(x) + f'(x_0)(x - x_0) + \dfrac{f''(x_0)(x - x_0)^2}{2!}$

$$+ \frac{f'''(x_0)(x - x_0)^3}{3!} + \ldots + \frac{f^{(n)}(x_0)(x - x_0)^n}{n!}$$

Third degree Taylor polynomial of $f(x)$ is $P_3(x) = f(x) + f'(x_0)(x - x_0)$

$$+ \frac{f''(x_0)(x - x_0)^2}{2!} + \frac{f'''(x_0)(x - x_0)^3}{3!}$$

1) $f(x) = \cos x$

Find a third-degree polynomial $P_3(x)$ centered at $c = \dfrac{\pi}{4}$

a) $f\left(\dfrac{\pi}{4}\right) =$?

b) $f'(x) =$? $\qquad f'\left(\dfrac{\pi}{4}\right) =$?

c) $f''(x) =$? $\qquad f''\left(\dfrac{\pi}{4}\right) =$?

d) $f'''(x) =$? $\qquad f'''\left(\dfrac{\pi}{4}\right) =$?

e) $P_3(x) = f\left(\dfrac{\pi}{4}\right) + f'\left(\dfrac{\pi}{4}\right)\left(x - \dfrac{\pi}{4}\right) + \dfrac{f''\left(\dfrac{\pi}{4}\right)\left(x - \dfrac{\pi}{4}\right)^2}{2!} + \dfrac{f'''\left(\dfrac{\pi}{4}\right)\left(x - \dfrac{\pi}{4}\right)^3}{3!} =$?

f) Draw graph of $f(x) = \cos x$ and $P_3(x)$ from $x = 0$ to $x = \pi$

2) $f(x) = \sin x$

Find a third-degree polynomial $P_3(x)$ centered at $c = \dfrac{\pi}{3}$

a) $f\left(\dfrac{\pi}{3}\right) =$?

b) $f'(x) =$? $\qquad f'\left(\dfrac{\pi}{3}\right) =$?

c) $f''(x) =$? $\qquad f''\left(\dfrac{\pi}{3}\right) =$?

d) $f'''(x) =$? $\qquad f'''\left(\dfrac{\pi}{3}\right) =$?

e) $P_3(x) = f\left(\dfrac{\pi}{3}\right) + f'\left(\dfrac{\pi}{3}\right)\left(x - \dfrac{\pi}{3}\right) + \dfrac{f''\left(\dfrac{\pi}{3}\right)\left(x - \dfrac{\pi}{3}\right)^2}{2!} + \dfrac{f'''\left(\dfrac{\pi}{3}\right)\left(x - \dfrac{\pi}{3}\right)^3}{3!} =$?

f) Draw graph of $f(x) = \sin x$ and $P_3(x)$ from $x = 0$ to $x = \pi$

3) $f(x) = x \cdot e^x$

Find a third-degree polynomial $P_3(x)$ centered at $c = 2$

a) $f(2) =$ ____?____

b) $f'(x) =$ ____?____ $f'(2) =$ ____?____

c) $f''(x) =$ ____?____ $f''(2) =$ ____?____

d) $f'''(x) =$ ____?____ $f'''(2) =$ ____?____

e) $P_3(x) = f(2) + f'(2)(x-2) + \dfrac{f''(2)(x-2)^2}{2!} + \dfrac{f'''(2)(x-2)^3}{3!} =$ ____?____

f) Draw graph of $f(x) = \sin x$ and $P_3(x)$ from $x = 0$ to $x = 4$

4) $f(x) = \ln x$

Find a third-degree polynomial $P_3(x)$ centered at $c = 2$

a) $f(2) =$?

b) $f'(x) =$? $\qquad f'(2) =$?

c) $f''(x) =$? $\qquad f''(2) =$?

d) $f'''(x) =$? $\qquad f'''(2) =$?

e) $P_3(x) = f(2) + f'(2)(x-2) + \dfrac{f''(2)(x-2)^2}{2!} + \dfrac{f'''(2)(x-2)^3}{3!} =$?

f) Draw graph of $f(x) = \ln x$ and $P_3(x)$ from $x = 0$ to $x = 4$

5) $f(x) = \dfrac{1}{x}$

Find a third-degree polynomial $P_3(x)$ centered at $c = 2$

a) $f(2) =$?

b) $f'(x) =$? $\qquad f'(2) =$?

c) $f''(x) =$? $\qquad f''(2) =$?

d) $f'''(x) =$? $\qquad f'''(2) =$?

e) $P_3(x) = f(2) + f'(2)(x-2) + \dfrac{f''(2)(x-2)^2}{2!} + \dfrac{f'''(2)(x-2)^3}{3!} =$?

f) Draw graph of $f(x) = \dfrac{1}{x}$ and $P_3(x)$ from $x = 0$ to $x = 4$